

Food Insecurity Forecast - Model Comparison

Two models compared: the original model and the enhanced model. Both are population-weighted ridge regression transition models trained on national ZCTA data from 2020-2024 and used to forecast Feeding Tampa Bay ZIP/ZCTA-level food insecurity rates for 2025-2027.

Short answer. The enhanced model is the better candidate based on the available validation test: its Feeding Tampa Bay average error is about **44% lower** than the original model. Both models predicted the 2024 holdout year too high on average, but the enhanced model was closer.

Summary SIDE-BY-SIDE

MODEL	MODEL TYPE	WHAT IT USES	VALIDATION RESULT	READ THIS AS
Original model	Population-weighted ridge regression transition model.	Prior-year food insecurity, unemployment, poverty, income, demographics, homeownership, disability, and a linear year trend.	FTB holdout MAE 0.0219; predicted 17.4% when actual was 15.4%.	Simple and transparent, but too high in the 2024 local test.
Enhanced model	Population-weighted ridge regression transition model.	Everything in the original model except the year trend, plus county food cost, county context, and ALICE hardship measures.	FTB holdout MAE 0.0123; predicted 16.2% when actual was 15.4%.	More accurate in the 2024 test, though still slightly high.

Lower MAE / RMSE is better. These errors are measured on a 0-1 rate scale, so 0.0123 means about 1.2 percentage points of average error.

How the Models Work

$\text{RATE}[T] \leftarrow \text{RATE}[T-1] + \text{DRIVERS}$

Both models use a transition-model structure. In plain English, each model starts with what food insecurity looked like last year, then adjusts the estimate using economic, demographic, and local-condition variables. For the forecast, the models are recursive: 2025 uses the observed 2024 rate, 2026 uses the predicted 2025 rate, and 2027 uses the predicted 2026 rate.

How to read the years. The models train on observed years **2020-2024**. The validation test hides 2024, trains on earlier transitions, then asks: how close would the model have come to the 2024 values we now know?

Projecting Future Inputs

DRIVERS FOR 2025-2027

Before either model can forecast food insecurity, it needs future values for the input drivers, such as unemployment, poverty, income, homeownership, disability, population, demographics, and, for the enhanced model, county food-cost and ALICE variables. Because independent future forecasts for every ZIP/ZCTA-county row are not included in the source data, the models create internal driver projections.

STEP	WHAT HAPPENS
Start with the latest observed row	The 2024 value is used as the baseline for each Feeding Tampa Bay ZIP/ZCTA-county row.
Estimate a recent trend	For each row and each driver, the model estimates a simple straight-line trend from the observed 2020-2024 history.
Extend that trend forward	The trend is carried forward into 2025-2027. For example, a driver that has been rising is projected to keep rising unless constrained.
Keep values in valid ranges	Rate-like inputs are kept between 0% and 95%, and projected population is not allowed to go below zero.
Use projected drivers in the food insecurity model	Each forecast year uses that year's projected drivers plus the prior-year food insecurity rate.

Original vs. enhanced input projections. The original model uses the simple linear driver projection with basic validity checks. The enhanced model uses the same linear projection, then adds stabilizing guardrails: key annual driver changes are capped, population is constrained to a broad +/-3% annual band around the 2024 baseline, and structural fields such as SNAP threshold, rural/urban code, and ALICE data availability are held constant unless new data is supplied.

Performance

FEEDING TAMPA BAY HOLDOUT

QUESTION	ORIGINAL MODEL	ENHANCED MODEL	WINNER
Average local miss	2.2 percentage points	1.2 percentage points	Enhanced
Typical large-error penalty	RMSE 0.0244	RMSE 0.0153	Enhanced
Bias direction	Predicted too high	Predicted too high	Enhanced is closer
Actual weighted 2024 rate	15.4%	15.4%	Same test
Predicted weighted 2024 rate	17.4%	16.2%	Closer: enhanced

The enhanced model also improves national 2024 holdout MAE from 0.0229 to 0.0151 and improves in-sample weighted MAE from 0.0122 to 0.0100.

What Changed in the Enhanced Model ADDED DRIVERS

The enhanced model adds variables meant to describe local cost pressure and household financial strain, not just ZIP-level demographics. This gives the model more context, but it also makes the model harder to explain and increases the chance of overlapping signals.

GROUP	ORIGINAL MODEL	ADDED IN ENHANCED MODEL	WHY IT MATTERS
Food insecurity history	Prior-year food insecurity rate	No change	Recent local history is usually the strongest signal.
Economic conditions	Unemployment, poverty, median income	No change	Captures financial stress and resources.
Demographics	Black share, Hispanic share, disability rate	County child share; county White non-Hispanic share	Adds broader county composition context.
Housing	Homeownership rate	No change	Often proxies for wealth and residential stability.
Food costs	Not included	Cost per meal; food cost index	Captures whether food is more expensive locally.
County context	Not included	SNAP threshold; rural/urban code	Adds policy and geography context.
ALICE hardship	Not included	Financial insecurity, poverty households, survival-budget thresholds, data flag	Adds household budget pressure measures.

Forecast Pattern 2025-2027

YEAR	ORIGINAL AVG RATE	ENHANCED AVG RATE	ENHANCED MIN-MAX	DIFFERENCE
2025	17.5%	17.0%	0.1% - 64.3%	-0.4 pts
2026	19.3%	19.2%	0.8% - 67.3%	-0.1 pts
2027	21.0%	21.7%	2.1% - 70.2%	+0.7 pts

Both outputs contain 259 Feeding Tampa Bay ZCTA/county rows per year. The enhanced model starts slightly lower than the original model in 2025, stays slightly lower in 2026, then ends higher by 2027.

Recommendation

MODEL USE

Recommended candidate: enhanced model. It is the better-performing model in the available holdout test, and it includes important local context around food costs and financial hardship. Use the original model as a benchmark because it is simpler and easier to explain.

NEXT STEP	WHY
Review or calibrate bias	The enhanced model predicted 2024 too high by about 0.8 percentage points on average for Feeding Tampa Bay rows.
Test a simpler enhanced model	Some enhanced-model variables overlap conceptually; a smaller version may be nearly as accurate and easier to defend.
Keep local validation separate	National performance is useful, but the decision audience cares most about the Feeding Tampa Bay region.
Show uncertainty in planning materials	Forecasts are directional estimates, not exact future counts.

Plain-English Glossary

TERMS USED ABOVE

TERM	MEANING
ZCTA	Census approximation of a ZIP Code area.
Holdout test	A test where the model predicts a year we already know, so we can measure its error.
MAE	Mean absolute error: the average size of the prediction miss. Lower is better.
RMSE	A second error measure that penalizes bigger misses more heavily. Lower is better.
Bias	Whether predictions tend to be too high or too low.
Ridge regression	A linear model with a stabilizing penalty so correlated variables do not swing the forecast too much.

Technical Appendix COEFFICIENTS

The models standardize predictors before fitting ridge regression. That means each coefficient reflects a one-standard-deviation change in the input, not a one-unit raw change. Treat these as predictive associations, not causal effects.

ORIGINAL MODEL DRIVER	COEF.	WHAT IT MEANS
Prior-year food insecurity rate	+0.0298	A ZIP/ZCTA's recent food insecurity history.
Unemployment rate	+0.0055	Local unemployment.
Poverty rate	+0.0125	Share of residents below poverty.
Percent Black	-0.0039	ZIP/ZCTA demographic composition.
Percent Hispanic	+0.0025	ZIP/ZCTA demographic composition.
Median income, log	-0.0021	Income level, log-scaled.
Homeownership rate	-0.0045	Share of homes owner-occupied.
Disability rate	+0.0073	Share of residents with a disability.
Centered year	+0.0085	Linear time trend in the original model.

ENHANCED MODEL DRIVER	COEF.	WHAT IT MEANS
Prior-year food insecurity rate	+0.0452	A ZIP/ZCTA's recent food insecurity history.
Unemployment rate	+0.0027	Local unemployment.
Poverty rate	+0.0064	Share of residents below poverty.
Percent Black	-0.0018	ZIP/ZCTA demographic composition.
Percent Hispanic	+0.0037	ZIP/ZCTA demographic composition.
Median income, log	+0.0036	Income level, log-scaled.
Homeownership rate	-0.0038	Share of homes owner-occupied.
Disability rate	+0.0054	Share of residents with a disability.
County child population share	-0.0002	Share of county residents who are children.
County % white, non-Hispanic	+0.0015	County demographic composition.
County cost per meal	+0.0218	Estimated average meal cost.

ENHANCED MODEL DRIVER	COEF.	WHAT IT MEANS
County food cost index	-0.0213	County food cost relative to national average.
County SNAP threshold	-0.0019	SNAP eligibility income threshold.
Rural-urban code	-0.0008	County rural/urban classification.
ALICE financial insecurity rate	-0.0001	Household hardship measure from ALICE data.
ALICE poverty household rate	+0.0003	Household poverty measure from ALICE data.
ALICE threshold under 65	+0.0009	Survival-budget threshold for under-65 households.
ALICE threshold 65+	+0.0001	Survival-budget threshold for 65+ households.
Has ALICE data	-0.0007	Flag for whether county ALICE data was available.

Forecast output sources were filtered to 2025-2027 for this report.